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ORIGINAL ARTICLE

Safety and Feasibility of Moderate-Intensity Home-Based Guided Exercise in Aortic Dissection Survivors in the United States: A Multicenter Randomized Controlled Trial

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BACKGROUND: Survivors of aortic dissection frequently restrict personal physical activity due to concerns about provoking recurrent aortic events. We conducted a multicenter randomized controlled trial to evaluate the safety and feasibility of a moderate-intensity, home-based guided exercise (GE) regimen.

METHODS: Between December 2022 and October 2024, adults ≥3 months post-Type A or B thoracic aortic dissection were randomized at 3 US academic medical centers to GE or usual care (UC; standardized exercise counseling and routine clinic visits). GE participants completed individualized training on a 6-exercise circuit and performed 12 months of home exercise with virtual follow-up visits. Coprimary outcomes were safety (aortic events, exertional hypertension, symptoms, injury, or events requiring medical care) and Patient-Reported Outcomes Measurement Information System 29-Item Profile scores. Secondary outcomes included exertional and ambulatory blood pressure and adherence. Continuous outcomes were compared using *t* or Mann-Whitney *U* tests; categorical using χ^2 or Fisher tests.

RESULTS: Ninety-three participants (mean age, 56 years; 30% women; 67% Type A aortic dissection) were randomized to GE (n=44) or UC (n=49). Sixteen participants (5 GE and 11 UC) were lost to follow-up, and 65 participants completed all trial milestones. There were no deaths, aortic operations, or recurrent dissections in either group. Exertional hypertension during supervised training occurred in 17 (39%) GE participants and was mitigated by exercise modification. There were no significant changes in paired ambulatory blood pressure measurements (25 GE, 28 UC) or Patient-Reported Outcomes Measurement Information System 29-Item Profile scores (30 GE, 35 UC).

CONCLUSIONS: In this pilot randomized trial, moderate-intensity home exercise appeared feasible and was not associated with observed excess aortic events compared with usual care. Larger prospective trials are needed to determine long-term effects on cardiovascular outcomes.

Key Words: adult ■ blood pressure ■ exercise ■ humans ■ middle aged

Survivors of thoracic aortic dissection (TAD) face lifelong risks for adverse aortic events, chronic hypertension, impaired functional capacity, and psychological distress. Although early surgical or endovascular repair improves survival, long-term outcomes remain strongly influenced by blood pressure control, physical activity patterns, and health-related quality of life.[1–3] Fear of precipitating recurrent dissection, particularly during physical exertion, frequently leads patients to restrict activity, limiting cardiometabolic and psychosocial recovery. Observational

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WHAT IS KNOWN

- Aortic dissection survivors are often counseled to restrict physical activity based on case reports of aortic events during high-intensity exertion, despite limited prospective safety data.
- No randomized trial has evaluated structured home-based exercise after aortic dissection, leaving exercise recommendations in this high-risk population guided largely by expert opinion.

WHAT THE STUDY ADDS

- In this pilot randomized trial, a 12-month moderate-intensity home-based guided exercise program was feasible and was not associated with deaths, aortic operations, or recurrent dissections compared with usual care.
- Exertional hypertension occurred in a minority of participants and was effectively mitigated through exercise modification.
- A fully powered trial with scalable digital monitoring tools is needed to evaluate long-term cardiovascular and psychosocial benefits.

Nonstandard Abbreviations and Acronyms

GE	guided exercise
PROMIS-29	Patient-Reported Outcomes Measurement Information System 29-Item Profile
TAD	thoracic aortic dissection
UC	usual care
UTH	University of Texas Health Science Center at Houston

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studies demonstrate that TAD survivors commonly receive inconsistent exercise guidance and often adopt sedentary lifestyles despite their elevated cardiovascular risk.[4–6]

Concerns about exercise-induced surges in systolic blood pressure (SBP) are derived from case reports of acute aortic syndromes that occurred during heavy weightlifting or high-intensity resistance training.[7–9] There is no evidence linking moderate-intensity physical activities to recurrent aortic dissection. In contrast, a large body of literature in nonaortic dissection populations shows that regular low-to-moderate intensity aerobic exercise improves central hemodynamics, reduces arterial stiffness, lowers blood pressure, and decreases major cardiovascular events and mortality in a dose-dependent manner.[10–12] Recent studies conducted using mouse models also support that low-to-moderate intensity exercise significantly reduces mortality and cardiac extracellular matrix degeneration while preserving aortic morphology and preventing pathological aortic dilation, supporting biological plausibility for benefit.[13,14]

Despite these potential advantages, participation in traditional center-based cardiac rehabilitation among aortic dissection survivors is extremely low, reflecting both structural barriers and uncertainty about safety.[15] Home-based cardiac rehabilitation has emerged as an effective alternative capable of delivering comparable improvements in functional capacity, adherence, and health-related quality of life while expanding access to medically vulnerable populations.[11,16] However, no randomized trial has evaluated the safety or efficacy of home-based exercise after aortic dissection, and professional guidelines, as well as other studies, acknowledge the absence of evidence to inform exercise prescriptions for this high-risk group.[1,17,18]

We conducted a pilot randomized controlled trial to evaluate the safety, hemodynamic responses, feasibility, and patient-reported outcomes associated with a structured, moderate-intensity home-based cardiac rehabilitation-style program for TAD survivors. We hypothesized that guided moderate-intensity home exercise would be safe without increasing adverse aortic events, improve mental health and health-related quality of life, and could improve blood pressure parameters compared with usual care (UC).

METHODS

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Ethics

The study protocol (HSC-MS-22-0936) was reviewed and approved by the Committee for the Protection of Human Subjects at the University of Texas Health Science Center at Houston (UTH) on November 28, 2022. Patients or the public were not involved in the design, conduct, analysis, or reporting of the trial. All participants signed electronic informed consent documents implemented in Research Electronic Data Capture before enrollment.[19]

Recruitment and Enrollment

Participants were identified by clinician referral and health record or database review at 3 sites: UTH, Washington University in St. Louis, and the University of Michigan. Potentially eligible patients were contacted by phone, approached in outpatient clinics, or recruited via social media.

Adults aged ≥18 years who had survived a Type A or Type B TAD at least 3 months before enrollment were eligible to participate in the trial. Participants were excluded if they were unable to attend 1 in-person exercise training session, had physical limitations preventing them from completing study exercises, could not access a treadmill or stationary bike at home, or had uncontrolled hypertension or symptomatic vascular, aortic, or coronary disease.

Trial Design

This multicenter, pragmatic, parallel-group randomized clinical trial (REGISTRATION: URL: https://www.clinicaltrials.gov;

Unique identifier: NCT05610462; registered November 9, 2022) was designed to evaluate preliminary efficacy and safety outcomes. The trial protocol and statistical analysis plan were previously published (Table S1).[20] After confirmation of eligibility and signed consent, participants completed 1 demographic survey and a baseline assessment using the Patient-Reported Outcomes Measurement Information System 29-Item Profile (PROMIS-29), v2.0 profile, which is validated to assess 7 health domains (physical function, fatigue, pain, depressive symptoms, anxiety, ability to participate in social roles and activities, and sleep disturbance).[21] Additional baseline assessments at enrollment included minutes engaged in and intensity of weekly exercise on the Behavioral Risk Factor Surveillance Survey, ambulatory blood pressure (using 24-hour ambulatory blood pressure monitors; OnTrak, Spacelabs, Inc), and arterial pressure waveform and pulse wave velocity (SphygmoCor XCEL, AtCor Medical, Inc).[22] At the conclusion of the study, participants completed the same assessments. Participants were then randomized 1:1 into guided exercise (GE) and UC groups. Participants were randomized using a stratified randomization scheme as implemented in Research Electronic Data Capture, with stratification by sex and study site, to ensure balance between study groups. A central computer at the primary study site (UTH) was used to implement the random allocation. None of the personnel who assigned participants to the intervention had access to the randomization sequence. Participants in both study arms continued to receive usual clinical care, including routine surveillance imaging, guideline-directed medical therapies, scheduled clinic visits, and any clinically indicated interventions, without input from or restrictions imposed by the study team (Figure 1).

Guided Exercise

Immediately after enrollment, participants in the GE group participated in a supervised exercise session, during which they completed 2 circuits of 6 different exercises at a moderate intensity level: hand grips, wall sits, stationary cycling, treadmill walking, bicep curls, and leg raises (Figure 2).

The in-person exercise training session was supervised by cardiac rehabilitation faculty and staff to ensure participant safety. Exercise, supervision, and blood pressure measurement methods were previously described.[23] Handgrip resistance was set at 40% of maximal effort using the dominant hand. Wall sits were completed at a 90-degree angle between the lower legs and back. Stationary cycling was completed at a 100-Watt target. Treadmill exercises were performed at a 14% grade incline at 3 miles per hour. During the supervised in-person session, bicep curls were performed with the dominant arm using 5-, 8-, or 10-pound weights to standardize the exercise and allow blood pressure measurement on the nondominant arm. For home sessions, participants were instructed to perform bicep curls using both arms. Participants completed leg raises by lifting both heels 6 inches off the ground in a supine position. A 24-hour ambulatory blood pressure monitor device was used to measure blood pressure during and after each exercise. Blood pressure measurements during exercise were initiated 15 seconds after the exercise began. Participants continued to exercise at least until the measurement was completed or until the exercise was performed at a steady intensity for 2 minutes.

Participants in the GE group were counseled individually on implementing the exercise circuit at home and provided with equipment for home use, with the exception of the treadmill and bicycle. They were advised to exercise at a moderate intensity for at least 150 minutes per week. They were encouraged to document other activities in exercise diaries, which were collected for review at follow-up visits.

GE participants participated in 3 virtual follow-up visits to assess progress and address participant questions. Virtual visits were conducted at 1, 3, and 9 months post-enrollment. The

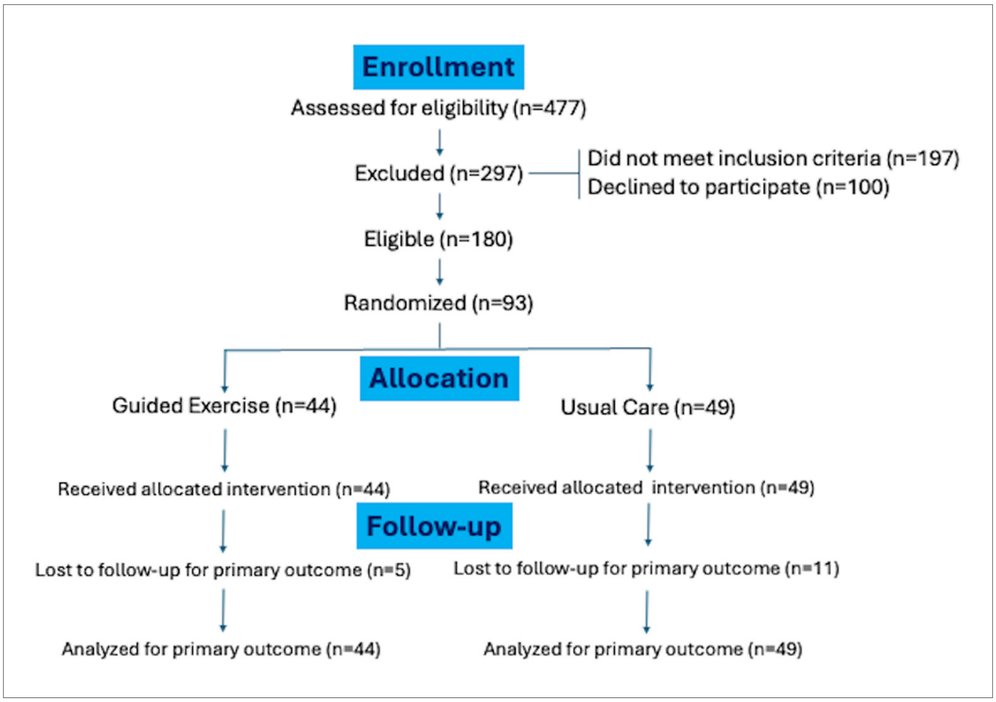

Figure 1. CONSORT flowchart showing participant screening, randomization, allocation follow-up, and analysis. Participants were followed for 12 months after enrollment. CONSORT indicates Consolidated Standards of Reporting Trials.

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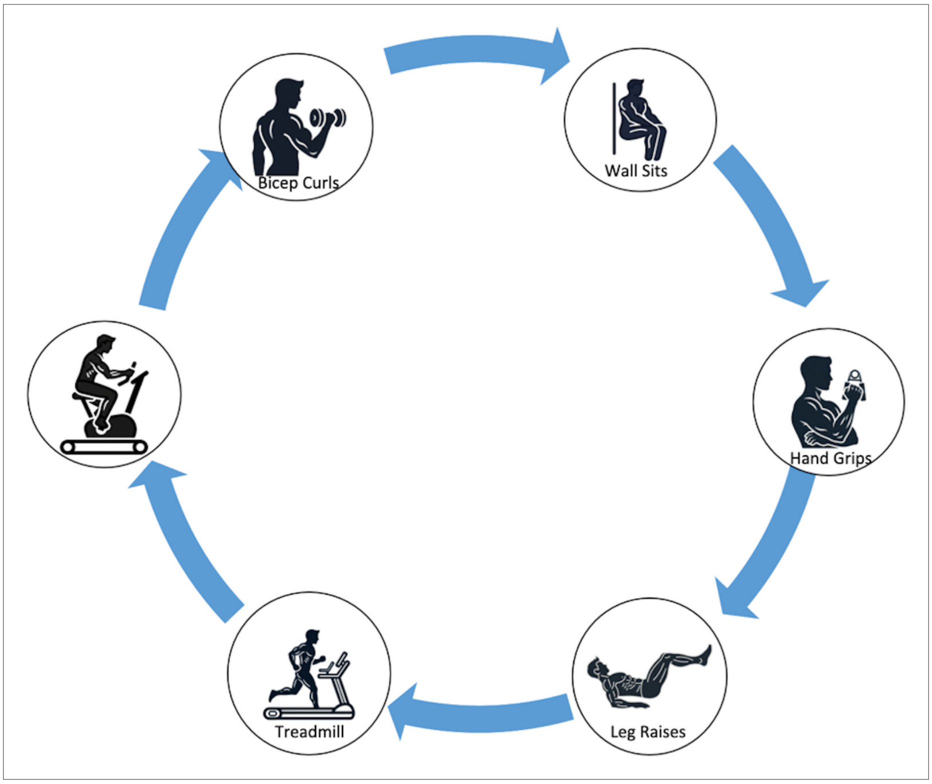

Figure 2. Illustration of the guided exercise circuit.
There was a 3-minute rest period between exercises. Blood pressure was measured once during each exercise and once during each rest period. Each participant completed the circuit twice.

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first virtual visit, conducted 1 month post-enrollment, consisted of participants demonstrating 1 exercise that study personnel could observe and correct if needed. At all virtual visits, participants answered questions regarding changes in their intensity of exercise through the Behavioral Risk Factor Surveillance Survey questionnaire, as well as questions related to updates in medication, exercise progression, clinic visits, and overall health status.

Usual Care

Participants who were randomized to the UC group completed a 24-hour ambulatory blood pressure monitor as part of the enrollment visit but did not participate in the in-person training session or virtual follow-up visits. UC participants were provided standard counseling about the potential benefits of exercise, including a handout with clinical and lifestyle guidance for patients with thoracic aortic disease, and took part in their regular clinic visits. Both GE and UC participants completed identical final study visits, which included a second 24-hour ambulatory blood pressure monitor measurement and completion of the Behavioral Risk Factor Surveillance Survey and PROMIS-29 questionnaires.

Primary Outcomes

The primary outcomes were participant safety and participant-reported quality of life and symptom burden as assessed by the PROMIS-29 Profile. The change in PROMIS-29 domain scores from baseline to postintervention constituted the primary efficacy end point. PROMIS T-scores were analyzed using previously reported methods.[21] A between-group or within-group difference of 5 T-score points was prespecified as the minimum clinically important difference. Assuming a 2-sided α of 0.05 and 80% power to detect a 5-point difference, a sample size of 63 participants per study arm (126 total) was required.

Participant safety was evaluated descriptively throughout the intervention period using a structured framework to capture exercise-related safety events. These included any untoward clinical occurrences temporally associated with the exercise intervention, such as exertional symptoms (dizziness, syncope, chest discomfort), hemodynamic abnormalities (eg, severe or sustained hypertension), musculoskeletal injuries, and any event requiring medical evaluation, emergency care, or study discontinuation. Exertional hypertension was defined as SBP >180 mm Hg or diastolic blood pressure >100 mm Hg during 2 or more exercises. Exercise was immediately discontinued if any of the following was noted: chest pain, significant fatigue or dyspnea, SBP >160 mm Hg after a recovery period of 3 minutes, any SBP >210 mm Hg or diastolic blood pressure >120 mm Hg, or a request to stop. There was no protocol-driven modification of antihypertensive therapies based on exertional hypertension observed during supervised exercise testing. Exertional hypertension was managed through exercise modification, and medication decisions were deferred to participants' treating clinicians as part of routine care. Exercise-related safety events were classified according to standardized criteria. Given the behavioral nature of the intervention, participants and intervention staff were not blinded; however, outcome adjudicators were blinded to group assignment, and standardized protocols were used across groups to ensure comparable assessment and management procedures.

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Secondary Outcomes

Exertional blood pressure response was assessed during standardized, supervised exercise testing sessions using a calibrated automated sphygmomanometer. Systolic and diastolic blood pressure were measured at rest, during predefined exercise stages, and during recovery, in accordance with American Heart Association–recommended measurement protocols. Secondary analyses evaluated peak exertional blood pressure, the rate of blood pressure increase with exercise, and the presence of exaggerated hypertensive responses using established clinical thresholds. Ambulatory blood pressure was measured using 24-hour ambulatory blood pressure monitoring performed at baseline and postintervention. Mean daytime, nighttime, and 24-hour systolic and diastolic blood pressure values were derived, along with measures of blood pressure variability and nocturnal dipping status. Attrition at virtual follow-up visits was a key secondary feasibility outcome and was defined as the proportion of enrolled participants who failed to complete ≥1 prespecified virtual follow-up assessments. Attrition rates were calculated at each follow-up time point and cumulatively across the study period, with reasons for missed visits or withdrawal documented when available. Participant engagement with the GE program was assessed secondarily using self-reported exercise logs. These data included the frequency and duration of completed exercise sessions and, when applicable, self-reported adherence to prescribed intensity ranges. Exercise log data were used descriptively and in exploratory analyses to contextualize attrition patterns and examine associations between engagement and changes in primary and secondary outcomes.

Statistical Analysis

Continuous variables were assessed for normality using visual inspection of distributions and the Shapiro-Wilk test. Associations between continuous variables were evaluated using Pearson or Spearman correlation coefficients. Normally distributed variables were compared using *t* tests for 2-group comparisons or 1-way ANOVA for multiple comparisons. When appropriate, post hoc pairwise comparisons were performed using Tukey adjustment. Nonnormally distributed continuous variables were compared using Mann-Whitney *U* tests or Kruskal-Wallis tests with Dunn post hoc testing as indicated. Categorical variables were compared using χ^2 or Fisher exact tests when expected cell counts were <5. Between-group differences in PROMIS outcomes were assessed using 2-sample comparisons of change scores; analyses using ANCOVA adjusting for baseline values yielded similar results. All statistical tests were 2-sided, and a $P<0.05$ was considered statistically significant. Analyses were performed using standard statistical software. All randomized participants were included in each analysis. Participants with missing outcome data were excluded from analyses involving those outcomes, and no statistical methods were used to address missingness.

RESULTS

A total of 477 potential participants were screened across the 3 study sites (247 at UTH, 64 at Washington University in St. Louis, and 166 at the University of Michigan), and 93 participants were eventually enrolled (45 at UTH, 20 at Washington University in St. Louis, and 29 at the University of Michigan) and randomized to GE (n=44) or UC (n=49; Figure 1). Although the prespecified target sample size was 126 participants, enrollment was closed after 93 participants were randomized because recruitment was slower than anticipated due to the rarity of aortic dissection, the requirement for an in-person training visit, and eligibility exclusions related to uncontrolled hypertension, physical limitations, symptomatic cardiovascular disease, and access to home exercise equipment. The first participant was enrolled on December 21, 2022, and the last participant completed follow-up on October 30, 2024. Approximately two-thirds of participants had survived a Type A aortic dissection. Baseline demographic and clinical characteristics (Table 1), PROMIS-29 scores, and daytime ambulatory blood pressure measures (Table 2) were similar between study groups; numerical differences in nocturnal ambulatory blood pressure parameters were not significant. Use of antihypertensive therapies was common in both study groups, with prevalent β-blocker use and a similar overall distribution of antihypertensive medication classes. During the 12-month follow-up interval, changes in daytime and nighttime ambulatory SBP were similar between the GE and UC groups (Figure 3). Distributions of individual-level changes demonstrated substantial overlap between groups, with no significant differences in the central tendency or dispersion favoring either intervention. These findings were consistent across both daytime and nighttime ambulatory SBP measures.

Exercise Adherence

GE participants were instructed to maintain home diaries detailing the time they spent on each exercise in the circuit. They documented monthly exercise time (in minutes), wall sit angle (in degrees), and handgrip strength (in pounds). More than half of the participants returned diaries detailing their timed activities over a mean reporting interval of 5 months. Median self-reported exercise duration per participant remained stable over time (Table 3). Several participants reported that they engaged in additional physical activities beyond the study exercises such as hiking, housework, or kayaking. To assess participant adherence and identify patterns in exercise modifications, we analyzed self-reported activities at each virtual follow-up visit. We categorized reported interruptions or modifications to the exercise regimen as physical limitations, incomplete participation, or interfering life events.

Fourteen GE participants (32%) reported making significant adaptations to the exercise regimen, defined as an intentional change in the type, intensity, frequency, or modality of the prescribed exercises (Table S2). The principal reasons for adaptation were physical limitations (86%) or interfering life events (50%). The most

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Table 1. Baseline Characteristics of Trial Participants

Characteristic	Guided exercise, n=44	Usual care, n=49
Age, y	56 (11)	57 (12)
Sex		
Female	15 (34%)	14 (29%)
Male	29 (66%)	35 (71%)
Dissection type		
Type A	31 (76%)	31 (65%)
Type B	10 (24%)	17 (35%)
Unknown	3	1
Intervention for dissection		
No intervention	5 (12%)	12 (24%)
Open repair	32 (74%)	32 (65%)
TEVAR	6 (14%)	5 (10%)
Unknown	1	0
Time since dissection, d	1098 (579–2014)	747 (425–1278)
Ethnicity		
Hispanic or Latino	2 (4.5%)	3 (6.1%)
Not Hispanic or Latino	42 (95%)	46 (94%)
Race		
American Indian/Alaska Native	0 (0%)	2 (4.1%)
Asian	2 (4.5%)	3 (6.1%)
Black	5 (11%)	7 (14%)
Native Hawaiian or Other Pacific Islander	0 (0%)	1 (2.0%)
Other	0 (0%)	4 (8.2%)
White	37 (84%)	32 (65%)
Height, inches	70.0 (3.8)	69.0 (4.0)
Weight, pounds	191 (169–220)	200 (175–234)
Calculated BMI	28.2 (24.8–31.6)	30.2 (26.5–32.9)
Site		
STL	11 (25%)	8 (16%)
UMI	12 (27%)	17 (35%)
UTH	21 (48%)	24 (49%)

Baseline demographic and clinical characteristics of participants who were randomized to the guided exercise and usual care groups. Variables are reported as mean (SD), median (interquartile range), or number (percentage). BMI indicates body mass index; STL, Washington University in St. Louis; TEVAR, thoracic endovascular aortic repair; UMI, University of Michigan; and UTH, University of Texas Health Science Center at Houston.

frequently modified exercises were treadmill walking (8 mentions), bicep curls (8 mentions), and biking (7 mentions).

Safety

During the in-person exercise training sessions, blood pressure was monitored frequently while participants completed the exercise circuit. Exertional hypertension, as defined by SBP >180 mm Hg or diastolic blood pressure >100 mm Hg during more than 1 exercise, occurred in 17 of 44 (39%) GE participants (Table S3). Four GE participants developed SBP >200 mm Hg during wall sits. The wall sit produced the highest mean SBP (149 mm Hg). Participants who developed exertional hypertension were counseled about exercise adaptations if they developed exertional hypertension. For example, 90° wall sits that induced exertional hypertension were adjusted to less acute angles until they could be performed without inducing exertional hypertension. In a single case, an in-person exercise session was postponed but was completed later after adjustment of antihypertensive therapies. All GE participants completed the exercise circuit.

Table 2. Baseline PROMIS-29 Components, Blood Pressure, and Antihypertensive Medication Use by Study Group

Characteristic	Guided exercise, n=44	Usual care, n=49
Anxiety T-score	52 (40–56)	51 (40–58)
Depression T-score	41 (41.0–52.3)	48.9 (41.0–52.2)
Physical function T-score	52.6 (45–56.9)	48.2 (43.4–56.9)
Fatigue T-score	49 (46–53)	49 (40–55)
Sleep disturbance T-score	51 (44–54)	45 (44–53)
Social T-score	54 (50–60)	52 (47–61)
Pain interference T-score	53 (42–56)	50 (42–56)
Physical health summary T-score	0.28 (−0.37 to 0.71)	−0.05 (−0.63 to 0.75)
Mental health summary T-score	0.15 (−0.19 to 0.60)	0.08 (−0.43 to 1.03)
Moderate activities per week	6 (2–7)	5 (3.7)
Vigorous activities per week	1 (0–3)	1 (0–4)
Mean daytime ambulatory SBP	121 (13)	123 (13)
Mean daytime ambulatory DBP	69 (8)	70 (9)
Mean nighttime ambulatory SBP	101 (93–115)	111 (102–117)
Mean nighttime ambulatory DBP	56 (53–63)	63 (56–69)
Nocturnal systolic dipping (%)	17 (12–22)	8 (4–15)
Nocturnal diastolic dipping (%)	19 (7)	11 (8)
Antihypertensive medications	2 (2–3)	2 (2–3)
β-Blocker	40/42 (95%)	43/47 (91%)
ACE inhibitor/ARB	28/42 (67%)	24/47 (51%)
Diuretic	12/42 (29%)	8/47 (17%)
Calcium channel blocker	13/42 (31%)	19/47 (40%)
Another vasodilator	4/42 (10%)	4/47 (9%)

Baseline values of the PROMIS-29 components and blood pressure were compared between study groups. Medication data were available for 42 of 44 guided exercise and 47 of 49 usual care participants. ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; DBP, diastolic blood pressure; PROMIS-29, Patient-Reported Outcomes Measurement Information System 29-Item Profile; and SBP, systolic blood pressure.

To further characterize blood pressure phenotypes associated with exercise responses, we compared ambulatory blood pressure parameters between participants who did and did not develop exertional hypertension during the supervised exercise session (Table 4). Participants with exertional hypertension exhibited higher peak daytime ambulatory SBP, whereas other ambulatory

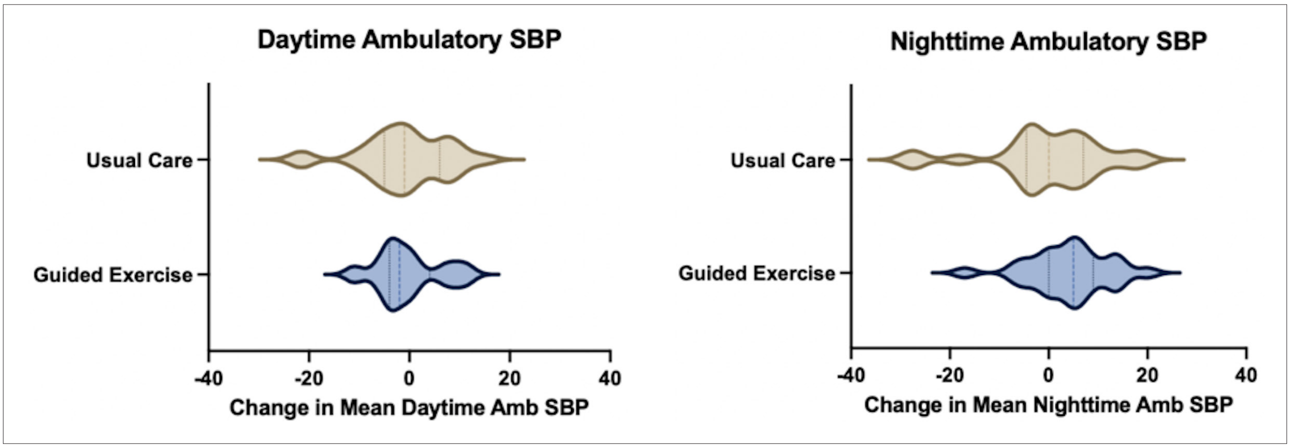

Figure 3. Distribution of changes in systolic blood pressure (SBP) over 12 months.
Violin plots illustrating the distribution of individual participant changes in SBP over 12 months in the guided exercise and usual care groups. Distributions are shown for daytime and nighttime ambulatory SBP. Vertical span of each graphic represents the relative density of observations.

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measures were not statistically different between the groups.

Patient-Reported Outcomes

There were no reported aortic operations, acute dissections, or participant deaths in the GE or UC groups. During the trial, a single GE participant experienced a right iliac artery dissection that was adjudicated as unrelated to the study intervention and attributed to activities outside the trial protocol (beach volleyball and cycling). The dissection remained stable on follow-up imaging and did not require intervention during the study period.

Paired PROMIS-29 measurements were available for 65 participants, including 30 GE and 35 UC participants; paired primary outcome data were missing for 28 of 93 (30%) participants, with similar missingness in the GE and UC groups. No PROMIS-29 domain demonstrated a statistically significant within-group or between-group change, and all observed changes were below the prespecified minimally clinically important difference of 5 T-score points (Table 5). The largest numerical between-group change was observed in sleep disturbance; the T-score of GE participants decreased by 1.2 points, whereas the T-score of UC participants increased by 0.8 points (*P*=0.28).

Table 3. Median Self-Reported Home Exercise Time (in Minutes) of Guided Exercise Participants

	Bicep curls	Wall sits	Hand grips	Leg raises	Treadmill running	Stationary cycling
Returned diary	25	32	30	31	27	28
Months logged per participant	5.0	5.3	5.2	5.2	4.9	5.0
Minutes per month	124.0	43.9	58.7	45.2	215.5	175.4

Returned diary: number of guided exercise participants (of 44) who returned exercise diaries containing entries for the specified exercise. Months logged per participant: mean number of months, among participants returning diaries for that exercise, in which exercise entries were recorded during the 12-month follow-up period. Minutes per month: median self-reported time spent on the specified exercise per month.

Attrition

An analysis of completed trial milestones demonstrated 16% attrition between virtual visits 1 and 2 and an additional 11% attrition between visits 2 and 3 (Figure S1). The trial completion rate varied by study site and was highest at UTH. Five participants in the GE group and 11 participants in the UC group were lost to follow-up after the initial enrollment visit.

DISCUSSION

In this multicenter randomized pilot trial evaluating structured home-based exercise after TAD, we found that a guided moderate-intensity program was feasible and was not associated with an observed excess of adverse aortic events during follow-up compared with UC. No deaths, recurrent dissections, or aortic surgeries occurred in either study arm, and exertional hypertension observed during supervised training was mitigated through individualized exercise modifications. These findings should be interpreted cautiously. The study was underpowered to detect modest but clinically meaningful differences, and type II error cannot be excluded. Accordingly, these results are best viewed as preliminary evidence supporting the feasibility of this strategy and informing the design of future clinical effectiveness trials.

Our findings challenge long-standing assumptions that TAD survivors should avoid moderate physical activity. Instead, our data demonstrate that with proper instruction and monitoring, TAD survivors can safely engage in

Table 4. Ambulatory Blood Pressure Data Stratified by Exertional Hypertension

Variable	Total (n=93)	Exertional HTN (n=17)	No exertional HTN (n=27)	*P* value
Mean SBP	118 (15)	122 (18)	114 (11)	0.11
Mean DBP	67 (11)	69 (10)	65 (6)	0.15
Day SBP	122 (18)	128 (17)	118 (14)	0.05
Day DBP	69 (12)	73 (14)	67 (6)	0.11
Night SBP	108 (18)	104 (24)	102 (14)	0.76
Night DBP	60 (13)	57 (11)	57 (5)	1.0
Peak daytime SBP	155 (26)	181 (37)	151 (20)	<0.01
Pulse pressure	51 (13)	54 (16)	48 (8)	0.17
Daytime SBP COV	11 (4)	14 (6)	11 (4)	0.08
Morning surge index (%)	16 (19)	19 (31)	16 (18)	0.72
Nocturnal dipping systolic (%)	12 (11)	16 (7)	15 (11)	0.71
Ambulatory Arterial Stiffness Index	0.48 (0.17)	0.52 (0.14)	0.52 (0.15)	1.0

Ambulatory blood pressure parameters for the overall cohort separated by the presence or absence of exertional HTN during supervised exercise. Values reported as mean (SD). Exertional hypertension was defined as SBP >180 mm Hg or DBP >100 mm Hg during >1 exercise. COV indicates coefficient of variation; DBPdiastolic blood pressure; HTN, hypertension; PP, pulse pressure; and SBP, systolic blood pressure.

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structured exercise without precipitating hypertensive crises or aortic complications. Importantly, peak daytime ambulatory SBP, rather than resting blood pressure, was associated with susceptibility to exertional hypertension, suggesting that ambulatory blood pressure phenotypes may help inform individualized exercise prescriptions.

This study has several important limitations. Enrollment did not reach the prespecified target sample size, and the trial was therefore underpowered to detect modest but clinically meaningful between-group differences in PROMIS-29 outcomes, ambulatory blood pressures, and safety comparisons. Baseline health-related quality-of-life T-scores were near population norms (≈50), creating a ceiling effect that limited the detectability of incremental improvement, and no PROMIS-29 domain demonstrated a statistically significant change in either arm over 12 months. Approximately 30% of participants lacked paired PROMIS-29 measurements, further reducing statistical power, although missingness was similar between groups. Multiple outcomes and within- and between-group comparisons were evaluated without a global multiplicity adjustment, increasing the possibility of type I error, while limited sample size may increase the likelihood of type II error; findings should therefore be interpreted with emphasis on overall patterns rather than individual *P* values. Recruitment was constrained by in-person visit requirements, and participants were highly selected and motivated, with many already engaged in regular physical activity, further contributing to ceiling effects and limiting generalizability. The cohort and trial steering committee were predominantly White and drawn from established clinical networks, which may limit broader applicability. Adherence was assessed using handwritten exercise diaries, which were incomplete and subject to reporting bias despite structured review with participants, and objective adherence measures and continuous physiological monitoring were not incorporated. Hemodynamic responses during home exercise were therefore not captured beyond supervised sessions and baseline/follow-up assessments. We also did not collect standardized longitudinal imaging data (eg, maximal aortic diameter or growth rates), detailed anatomic classifications of the dissections, or comprehensive cardiometabolic risk factor data, limiting assessment of structural aortic outcomes and broader metabolic effects. Genetic aortopathy status was not captured at enrollment, limiting the potential generalizability of our findings to heritable thoracic aortic diseases. Despite these limitations, qualitative participant feedback and adherence patterns suggest that structured exercise was feasible and could be incorporated into daily life with self-monitoring and modification. The absence of

Table 5. PROMIS-29 T-Scores at Baseline and Poststudy With Between-Group Analysis

Outcome	GE (n=30) baseline	GE post	GE change	UC (n=35) baseline	UC post	UC change	Mean difference (95% CI)*	Cohen *d*
Physical function	50.2 (8.3)	50.9 (7.2)	0.8 (8.7)	49.9 (7.1)	49.2 (8.1)	−0.8 (5.8)	1.6 (−2.1 to 5.3)	0.22
Anxiety	51.3 (9.8)	50.8 (8.7)	−0.5 (12.0)	49.2 (8.8)	48.8 (7.9)	−0.5 (8.3)	0.0 (−5.1 to 5.1)	0.00
Depression	45.8 (8.6)	46.3 (8.4)	0.6 (10.9)	46.6 (6.8)	46.9 (7.7)	0.2 (7.2)	0.4 (−4.0 to 4.8)	0.05
Fatigue	49.8 (8.8)	50.0 (9.4)	0.2 (8.2)	47.4 (10.3)	48.4 (9.0)	1.1 (7.9)	−0.9 (−4.9 to 3.1)	−0.11
Sleep disturbance	48.6 (7.0)	47.4 (7.6)	−1.2 (7.0)	47.0 (7.7)	47.8 (6.2)	0.8 (7.9)	−2.0 (−5.7 to 1.7)	−0.27
Social	54.4 (8.5)	55.1 (7.4)	0.7 (11.0)	53.6 (7.9)	53.3 (9.3)	−0.2 (8.5)	0.9 (−4.0 to 5.8)	0.11
Pain interference	50.1 (6.5)	50.6 (8.2)	0.5 (7.1)	50.1 (7.9)	49.2 (8.5)	−0.9 (8.0)	1.4 (−2.5 to 5.3)	0.19
Physical health summary	50.7 (7.9)	51.4 (7.0)	0.7 (8.4)	50.4 (7.2)	49.7 (8.5)	−0.7 (5.9)	1.4 (−2.1 to 4.9)	0.21
Mental health summary	51.8 (7.2)	52.0 (7.0)	0.2 (7.7)	52.9 (7.7)	52.3 (7.3)	−0.6 (6.0)	0.8 (−2.4 to 4.0)	0.12

Each value is reported as mean (SD). GE indicates guided exercise; and UC, usual care.

*Mean difference represents between-group difference in mean change (guided exercise−usual care). Mean differences and CIs were calculated using Welch 2-sample *t* tests based on observed change scores. Effect sizes (Cohen *d*) were calculated using pooled SD of change scores.

deaths, recurrent aortic dissections, or aortic reinterventions during follow-up in this selected cohort, including many participants with high-risk features such as prior Type A dissection and observed exertional hypertension, supports the safety of this approach.

The study cohort was enriched for survivors of Type A dissection who had undergone open repair, although a meaningful proportion had Type B dissections and had not undergone operative or endovascular interventions. While repaired Type A dissection is often perceived as lower risk, these patients remain susceptible to progressive distal aortic disease and late complications, and prospective data to guide exercise recommendations remain limited. At the same time, patients with medically managed, unrepaired dissections may represent a higher-risk subgroup in whom exercise safety is even less well established. Because the study was not powered for subgroup analyses, we cannot draw definitive conclusions across dissection phenotypes. Future studies should stratify participants by dissection type, repair status, and residual aortic pathology.

Our findings highlight the feasibility of home-based exercise programs but also underscore logistical barriers to conducting multicenter behavioral trials in rare disease populations. Larger, adequately powered pragmatic trials are warranted to evaluate long-term clinical outcomes, including recurrent aortic dissection, aortic dilation, and cardiovascular mortality, and should include comprehensive phenotyping of dissection subtype, repair status, residual aortic pathology, genetic background, and cardiometabolic risk factors to enable more precise, individualized risk assessment. Future trials may also benefit from incorporating wearable physiological monitoring, app-based data capture, objective adherence measures, and standardized imaging with centralized adjudication to reduce participant burden and improve data completeness, possibly including mechanistic end points such as central aortic stiffness, pulse wave velocity, and biomarkers of vascular stress. Precision exercise prescriptions tailored to ambulatory hemodynamic profiles with integration of scalable digital health tools may further enhance data quality and optimize exercise safety and adherence in this medically vulnerable population.

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Disclosures

None.

Supplemental Material

Tables S1–S3
Figure S1

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